

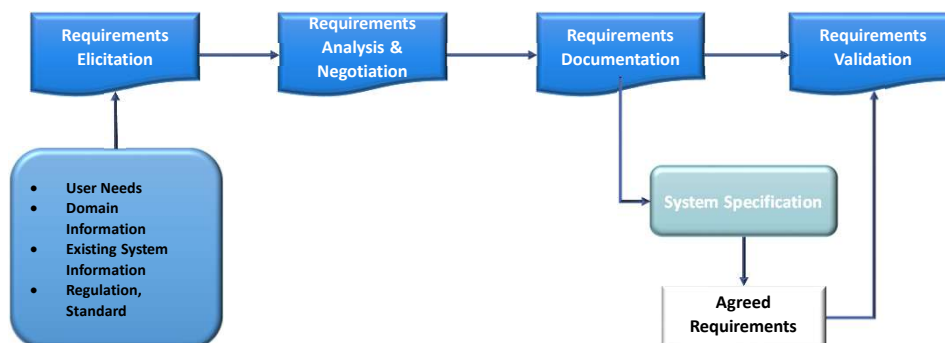
BUSINESS ANALYSIS

Lecture 7: Requirements Elicitation

1

THE PROCESS OF REQUIREMENTS ENGINEERING

Linear Requirements Engineering Model



2



REQUIREMENTS ELICITATION (RE)

- **What is Requirements Elicitation**

“Requirements elicitation (RE) is the initial step of requirements engineering where requirements analysts elicit information about the problem domain from available choices.” (Anwar & Razali, 2012)

- **RE Techniques: Some Broad Categories**

Traditional: Interviews, surveys, questionnaires

Collaborative: Focus groups, workshops, brainstorming

Prototyping: Simplified version of a software

Modelling: Scenarios, business process models

Agile: Mind mapping, user stories

References:

Anwar & Razali, 2012; A Practical Guide to Requirements Elicitation Techniques Selection - An Empirical Study

Pacheco et. al., 2018; Requirements elicitation techniques: a systematic literature review based on the maturity of the techniques

3



REQUIREMENTS ELICITATION: ANOTHER CATEGORIZATION SCHEME

- **Conversational:** Interviews, workshops/focus groups, brainstorming

Allow analysts to interact with stakeholders in directly and face to face conversations

- **Observational:** Ethnography, protocol analysis

Analysts spend time on-site monitoring and analyzing how stakeholders work

- **Analytical:** Requirements re-use, documentation

Analysts extract the required information from documentation or code of existing systems

- **Synthetic:** JAD or Joint Application Development, prototype, scenarios/storyboards

Combine conversational, observational and analytical approaches

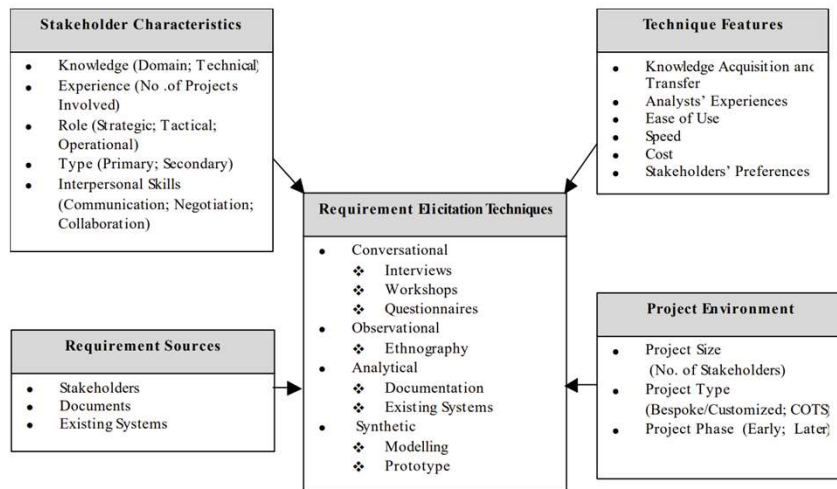
References:

Anwar & Razali, 2012; A Practical Guide to Requirements Elicitation Techniques Selection - An Empirical Study

Pacheco et. al., 2018; Requirements elicitation techniques: a systematic literature review based on the maturity of the techniques

4

FACTORS OF RE TECHNIQUES SELECTION



References: Anwar & Razali, 2012; A Practical Guide to Requirements Elicitation Techniques Selection - An Empirical Study

5

COMMON REQUIREMENT ELICITATION METHODS

- Interviews
- Workshops
- Observation
- Surveys
- Brainstorming

6



COMMON REQUIREMENT ELICITATION METHODS

- **Interviews**

elicit information from a person (or a group of people) in an informal or formal setting by asking questions and documenting/ recording responses

- **Types of interviews**

- Structured: Questions are pre-prepared
- Unstructured: Questions are not prepared, but evolve out of the responses of the interviewee
- Semi-structured: Combination of structured and unstructured technique

- **Main stages could involve**

- Planning
- Execution of the interview
- Follow-up

- **Interview mistakes**

Video: <https://www.youtube.com/watch?v=yBYd5USasXg>

Sources:

Business Analyst Mentor; https://businessanalystmentor.com/elicitation-technique/#elementor-toc_heading-anchor-11

Formplus; <https://www.formpl.us/blog/survey-methods>

7



COMMON REQUIREMENT ELICITATION METHODS

- **Workshops**

A structured meeting attended by multiple stakeholders interaction among whom is facilitated by the analyst or professional facilitator, which may conclude with the development of a model or prototype

- **Main stages could involve**

- Planning: e.g., selection of participants, date, location
- Opening: Communicating the purpose and agenda
- Execution: Facilitate the workshop and ensure that the purpose and agenda are not lost; conflicts are managed etc.
- Closing: Discuss the outcome, decision, achievements, and also what remained unachieved
- Follow up: Circulation of meeting notes, pictures, seeking feedback on the meeting etc.

- **Disadvantages of techniques such as brainstorming, workshops, or focus groups that involve a large group**

- Participants may not communicate their true thoughts and experiences thereby affecting the validity of the process.
- Herd opinion; participants may be influenced by the opinions of other people in the group or an influential or vocal participant

Sources:

Business Analyst Mentor; https://businessanalystmentor.com/elicitation-technique/#elementor-toc_heading-anchor-11

Formplus; <https://www.formpl.us/blog/survey-methods>

8



COMMON REQUIREMENT ELICITATION METHODS

- **Observation**

involves observing a business in person and trying to understand how a user does their job, or “job shadow”

- **Observation techniques**

- Communication with the people who are involved in a job or process and asking questions
- Documenting what is being done
- Documenting observations and planning follow-up questions

- **Disadvantage**

Participants may act differently when aware of being observed, thereby affecting the accuracy of observations

Sources:

Business Analyst Mentor; https://businessanalystmentor.com/elicitation-technique/#elementor-toc_heading-anchor-11

Formplus; <https://www.formpl.us/blog/survey-methods>

9



COMMON REQUIREMENT ELICITATION METHODS

- **Surveys**

is mostly a set of structured questions to which respondents provide answers based on their knowledge and experiences. Surveys may be used when there are many stakeholders or when stakeholders are geographically dispersed, and you need to gather the same information from them

- **Examples of survey techniques**

- Household surveys
- Mail surveys
- Online questionnaires

- **Disadvantages**

- Survey response bias.
- High survey dropout rates can also affect the number of responses

Sources:

Business Analyst Mentor; https://businessanalystmentor.com/elicitation-technique/#elementor-toc_heading-anchor-11

Formplus; <https://www.formpl.us/blog/survey-methods>

10



COMMON REQUIREMENT ELICITATION METHODS

- **Brainstorming**

involves spontaneous generation of ideas, often in a larger group of people with a white board or some other mechanism to capture the ideas.

- **Brainstorming technique**

- Arrange a meeting / workshop of the participants
- Explain the purpose, agenda, scope
- Initiate the session with an idea or proposal
- Invite participants to contribute ideas and build on others' ideas.
- Facilitate the discussion
- Take notes

- **Disadvantages of techniques such as brainstorming, workshops, or focus groups that involve a large group**

- Participants may not communicate their true thoughts and experiences thereby affecting the validity of the process.
- Herd opinion; participants may be influenced by the opinions of other people in the group or an influential or vocal participant

- **Brainstorming for the best results**

To get the best out of the session let human imaginations fly and DO NOT dismiss an idea as preposterous or ridiculous

Sources:

Business Analyst Mentor; https://businessanalystmentor.com/elicitation-technique/#elementor-toc_heading-anchor-11

Formplus; <https://www.formpl.us/blog/survey-methods>

11



GENERAL GUIDELINES ON ELICITATION METHODS

- **Planning**

Planning without identifying stakeholders, or without focus on the objectives of elicitation can defeat the purpose of RE

- **Avoiding bias**

Design RE objective and avoid cherry-picking information that suits your personal ideas

- **Active listening**

Do not just "hear" but understand

- **Smart questions**

Plan and questions that are relevant to the analysis

Sources:

Business Analyst Mentor; https://businessanalystmentor.com/elicitation-technique/#elementor-toc_heading-anchor-11

Formplus; <https://www.formpl.us/blog/survey-methods>

12



ESSENTIAL READINGS

Anwar & Razali, 2012; A Practical Guide to Requirements Elicitation Techniques Selection - An Empirical Study

Business Analyst Mentor

Page title: 9 Elicitation Techniques Used by Business Analysts

Available at: https://businessanalystmentor.com/elicitation-technique/#elementor-toc_heading-anchor-11

Formplus

Page title: Survey methods: Definition, types, and examples

Available at: <https://www.formpl.us/blog/survey-methods>

Pacheco et. al., 2018; Requirements elicitation techniques: a systematic literature review based on the maturity of the techniques

13



REFERENCE BOOKS OPTIONS

- **BUSINESS ANALYSIS BODY OF KNOWLEDGE (BABOK);** IIBA Bookstore: <https://www.iiba.org/career-resources/a-business-analysis-professionals-foundation-for-success/babok/>
- **BUSINESS ANALYSIS (Fourth Edition)** by Debra Paul (Author), James Cadle (Author), Malcolm Eva (Author), Craig Rollason (Author), Jonathan Hunsley (Author)
- **BUSINESS ANALYSIS TECHNIQUES: 123 ESSENTIAL TOOLS FOR SUCCESS (3RD Edition)** by James Cadle (Author), Debra Paul (Author), Jonathan Hunsley (Author), Adrian Reed (Author)

14